

ITU

ICTs and Climate Change

Meeting with UNFCCC
6 May 2010

Committed to connecting the world

Introduction to ITU

- Founded in 1865, oldest specialised agency of the UN
- Founded to ensure interoperability of international communications
- Remains key function of ITU
- 191 Member States, 780 private sector entities
- HQ Geneva, 11 regional offices, 760 staff / 80 nationalities
- Named as one of the world's ten most enduring institutions by Booz Allen
- Five elected officials:
 - Secretary-General
 - Deputy Secretary-General
 - Director of the Radio Bureau (BR)
 - Director of the Telecommunication Standardization Bureau (TSB)
 - Director of the Telecommunication Development Bureau (BDT)

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ITU-T Standards (Recommendations) connect the world...

- Over 3000 standards
- Basis for the international telecommunications networks
- Increasing extending to all aspects of ICTs
- Without ITU standards the Internet wouldn't function.



Not all standards are equal

- Recommendations become mandatory if adopted in law
- Private standards may confuse users and consumers and do not ensure interoperability
- ITU's broad range of stakeholders, and robust processes provide the basis for consensus across sectors and countries
- Market-driven international standards, based on objective information and knowledge
- Meet the needs and concerns of all relevant stakeholders

Strategic Objectives

1. Bridge the digital divide
2. Build on broadband
3. Manage the spectrum and geostationary satellite orbit
4. Develop and publish timely global standards
5. Identify relevant areas for future standardization projects
6. Disseminate information and know-how
7. Capacity building
8. Projects to support and assist the membership, in particular developing countries

Key Features

- Open, transparent, consensus based, fast working, public/private partnership
- Technical standards developed by industry members, when consensus placed on website and if no comments after 4 weeks is in effect approved by 191 governments
- ITU standards are therefore truly global, open standards, unlike those of many other standards bodies, forums or consortium that claim to produce global and open standards, available free of charge
- Publicly available database of products and services meeting ITU standards
- Organizing interoperability events to prove interoperability of different vendors equipment
- Common IPR policy with ISO and IEC (FRAN)



Importance of Global Standards

- Global Standards essential in a complex world
- Standards make things easier
- Essential for international communications and global trade
- Drive competitiveness, for individual businesses and world economy
- Help organizations with their efficiency, effectiveness, responsiveness and innovation
- Lower prices and increase availability by reducing technical barriers and promoting compatibility between systems and networks
- Manufacturers, network operators and consumers benefit

ITU and Climate Change"



"Climate Change is a global challenge that the world cannot lose".

Dr Hamadoun I. Touré
ITU Secretary-General, 12 November 2008



"Climate change is the defining challenge of our era. ITU's work to cut greenhouse gas emissions, develop standards and use 'e-environment' systems can speed up the global shift to a low-carbon economy".

Ban Ki-moon
United Nations Secretary-General, 12 November 2008



**UN Secretary-General,
Ban Ki-moon: "ITU is one of the very
important stakeholders in the area of
climate change."**

- Methodology to describe and estimate present and future user [energy] consumption of ICTs over their entire life-cycle
- Smarter standards for greener systems & services



Data Monitoring

- **ITU facilitates climate monitoring:**
 - Conducting and managing studies on remote-sensing needs
 - Providing key climate data via radio-based applications
 - Active monitoring of key climatic variables
 - Close collaboration with WMO



Adaptation

- Support to get telecoms up and running after disasters
 - Recent examples Haiti and Chile
- E-Environment Toolkit will help countries to assess the contribution that ICTs can make to reduce GHG emissions
- Common alerting protocol
- Common number allocated to UNOCHR



Mitigation

- Reducing energy consumption
- The promotion of Next Generation Networks (reducing power consumption by up to 40%)
- Recycling, e-waste, lifecycle analysis
- All new standards are now checked for energy efficiency
- ITU-T Study Group 5 on Environment and Climate Change

ITU-T Recommendation L.1000: Universal charger

Delivers 50% reduction in standby energy consumption, will eliminate up to 82,000 tonnes of redundant chargers, and cuts GHG emissions by at least 13.6 million tonnes annually



New Technology

- Identifying priority sectors where ICTs can reduce emissions
 - Smart grid (new Focus Group)
 - Smart buildings
 - Intelligent transport systems
 - Remote working technologies
 - Sensor-based networks



Common Methodology

- Common methodology for measuring ICT carbon footprint
- Without, it will be impossible to provide meaningful comparisons
- Helps to establish the business case to go green
- Over 20 organisations participate in the ITU group



Joint UNFCCC/ITU Project?

- The Environmental Protection Agency (EPA) in Ghana has opened a dialogue with Vodafone Ghana
 - Review Vodafone Ghana's operation to assess environmental impacts and ways and contribute to appropriate mitigation actions.

Awareness Raising

- TECHWATCH Reports on CC and positive impact of new technologies
 - Next Generation Networks, Intelligent Transport Systems, etc.
- Major Symposia on ICTs and CC
 - 2008: Kyoto and London
 - 2009: Quito and Seoul (virtual event)
 - 2010: Egypt
- ITU-T pioneering energy efficient work methods
 - Paperless meetings, on-line work tools, virtual symposia
- ITU-T leading Dynamic Coalition on Internet and Climate Change as part of the Internet Governance Forum



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